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RESEARCH ARTICLE

Enhanced YOLOv8 Framework for Precise Small Object Detection in UAV Imagery

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ABSTRACT The rapid advancement of UAV technology is accelerating the intelligent transformation of object detection methods. UAV applications now span both civilian and military sectors, playing crucial roles in scenarios such as aerial surveying, disaster response, and security monitoring. However, UAV image detection faces numerous challenges: targets are affected by flight altitude and viewing angle, leading to significant scale variations; dynamic flight results in constantly changing spatial layouts and complex backgrounds—such as dense vegetation and building textures—often weaken or obscure object features, making detection more difficult. To tackle these issues, we present a novel YOLOv8-based model tailored for small object detection in UAV images. In response to the prevalence of small targets in UAV images, the model enhances the multi-scale network structure and introduces the Large Separable Kernel Attention-enhanced Spatial Pyramid Pooling Fast Layer (SPPF-LSKA), improving the model's capability to extract features from and respond to small objects. Furthermore, the introduction of the Minimum Point Distance IoU (MPDIoU) and Soft Non-Maximum Suppression (Soft-NMS) enables effective detection of overlapping and occluded targets. Experimental results on the VisDrone2019 dataset show that, compared with YOLOv8, the improved model increases mAP@0.5 by 7.6%, reduces parameter count by 23%, and has a model size of just 5MB. Compared with other current algorithms, the enhanced model delivers the best detection performance, highlighting the effectiveness of the proposed approach.

INDEX TERMS Object detection, UAV images, YOLOv8, feature extraction.

I. INTRODUCTION

In recent years, as the critical role of UAV image-based object detection in intelligent transportation systems, disaster response, and surveillance applications has become increasingly prominent, this field has emerged as a significant area of research [1], [2]. With the rapid advancement of deep learning, numerous object detection algorithms have been developed to address challenges such as scale variation, background complexity, and object occlusion in aerial scenes. These algorithms are broadly categorized into two-stage and single-stage detectors.

In the domain of aerial imagery, two-stage methods have shown considerable accuracy. Yang et al. [3] proposed a Gaussian heatmap-based label assignment for arbitrary-oriented object detection, which models target confidence using Gaussian distributions and integrates rotated RoI operations to enhance detection precision. Koyun et al. [4] employed a two-stage CNN framework with region proposal networks (RPN) and optimized RoI alignment, achieving high accuracy on datasets like VisDrone, especially for small targets. Munjal et al. [5] further improved small object detection through a "Focus-and-Detect" framework, leveraging attention mechanisms and multi-scale feature enhancement.

On the other hand, single-stage detection frameworks are favored for their real-time performance. Saini et al. [6] introduced SOLARNet, a lightweight regression-based network with adaptive receptive fields tailored for aerial imagery. Zhang et al. [7] proposed Adaptive Slicing-Aided Hyper Inference (ASHI) to boost small object detection in high-resolution remote sensing imagery, efficiently balancing accuracy and computational cost.

Complementing these detection-focused methods, recent advancements in remote sensing image processing have significantly contributed to the quality and interpretability

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of UAV imagery. For instance, Wang et al. [8] proposed VOGTNet, which employs a variational optimization-guided two-stage network to enhance the spatial and spectral consistency in multispectral and panchromatic image fusion. This provides higher-quality inputs for downstream tasks such as object detection. He et al. [9] introduced IGroupSS-Mamba, which incorporates a spatial–spectral attention mechanism for hyperspectral image classification, emphasizing the importance of fine-grained feature extraction in remote sensing imagery. Liu et al. [10] proposed MT_GAN, a GAN-based framework that performs SAR-to-optical image translation for cloud removal, demonstrating the potential of generative models in image reconstruction and enhancement.

Inspired by these works, we propose an enhanced YOLOv8-based framework tailored to the demands of small object detection in UAV imagery. Building upon the efficiency of single-stage detectors, we redesigned the multi-scale network architecture and incorporated an efficient attention mechanism and loss function to enhance detection accuracy with minimal compromise to real-time performance.

Although the YOLO algorithm has achieved great success in detecting objects in ground-level images, it still faces significant challenges when applied to UAV imagery. As shown in Figure 1 (a)–(d), UAV-captured images are affected by altitude and camera angle, often resulting in object overlap within the image. Figure 1 (a) illustrates the uneven spatial distribution of targets under an oblique UAV view, where vehicles and pedestrians overlap and dense crowds are present outside a supermarket, while the adjacent street remains relatively sparse. Figure 1 (b) demonstrates how pedestrians appear extremely small and nearly indiscernible to the naked eye when the UAV is flying at a high altitude. Figure 1 (c) shows blurred pedestrians captured at night, with visibility further affected by shadows. Figure 1 (d) depicts occlusion caused by trees and tall buildings, making the obstructed objects difficult to detect. These characteristics of UAV imagery pose considerable challenges to the stability and robustness of object detection algorithms.

In addition, UAV-based object detection algorithms typically have specific requirements regarding model size. On one hand, UAVs need to capture and store images; on the other hand, they must process these images for detection, which necessitates using object detection models that are as compact and efficient as possible to ensure real-time performance. However, with continuous technological advancements, improvements in storage capacity and computational speed have enabled UAVs to possess a certain degree of onboard storage and processing power. Therefore, the primary focus has shifted to enhancing detection accuracy. At the same time, it remains essential to select algorithms that require low computational resources and have a small model size for effective deployment in UAV-based object detection tasks.

To address the specific challenges of object detection in UAV imagery, this study introduces an enhanced detection model built upon YOLOv8n. Evaluated on the widely used VisDrone2019 dataset [11], the proposed model demonstrates improved detection accuracy while maintaining a relatively low parameter count and computational load. It significantly outperforms the baseline YOLOv8n in overall performance.

The main contributions of this paper are as follows:

(1) Innovative redesign of the feature pyramid structure: A high-resolution feature layer of size 160×160 is introduced, replacing the traditional minimum resolution of 80×80 in YOLOv8. This design significantly enhances the spatial detail retention capability of shallow layers, enabling the model to capture pixel-level fine-grained features such as small object edges and textures. Experimental results show that this improvement effectively mitigates the issue of small object feature loss caused by downsampling in UAV imagery, achieving a 3.1% increase in small object recall and a 2.3% improvement in mAP@0.5, while reducing the number of parameters by 23%.

(2) Large-kernel attention-enhanced feature extraction mechanism: A large kernel depthwise separable convolution attention module is integrated into the Spatial Pyramid Pooling-Fast (SPPF) block, which significantly enhances the multi-scale feature extraction capability of the SPPF module. When combined with the 160 × 160 feature map, the model demonstrates superior feature extraction performance for small objects.

(3) Detection optimization strategy for dense scenes: To address the challenges of heavy overlaps and occlusions commonly found in UAV imagery, a novel combination of Soft-NMS and MPDIoU is proposed. Soft-NMS replaces hard-threshold suppression with a Gaussian decay strategy, effectively reducing the miss rate for overlapping objects. MPDIoU, a loss function based on the minimum point distance, jointly optimizes for overlapping regions, center point distances, and width-height deviations for the first time. The integration of both methods significantly improves detection accuracy in overlapping areas, with mAP@0.5 increasing by 6%.

The remainder of this paper is structured as follows: Chapter 2 reviews recent advancements in object detection for UAV imagery, with an emphasis on the YOLO family of models. Chapter 3 details the proposed methodology developed to tackle UAV-specific detection challenges. Chapter 4 introduces the dataset, experimental design, evaluation criteria, ablation analysis, and comparative results. Chapter 5 explains how the proposed model addresses the challenges present in UAV imagery. Chapter 6 summarizes the contributions of this study, discusses its limitations, and outlines potential directions for future research.

II. RELATED WORK

In recent years, the rapid development of deep learning technology has brought revolutionary changes to the field of object detection. Whether based on two-stage or single-stage frameworks, object detection algorithms have achieved considerable success in the field. However, UAV imagery

(a) Overhead angle (b) Small object

(c) Motion blur (d) Occlusion

FIGURE 1. **Partial UAV image samples from VisDrone2019 dataset.**

presents far greater challenges than conventional object detection tasks due to its typically small image size, low resolution, dense object distribution, and complex backgrounds. As a result, UAV image detection has become a prominent research focus within the academic community. In recent years, researchers have proposed a variety of improved methods to address these challenges.

To address the issue of dense object clustering in UAV imagery, In 2019, Yang et al. [12] proposed ClusDet, a multi-stage cluster detection network designed to improve both efficiency and accuracy in high-resolution UAV image detection. This method employs a region clustering preprocessing step to segment potential object areas, combines a slice-based detection strategy to optimise computational load, and introduces a scale-adaptive mechanism to dynamically adjust the granularity of feature extraction. As a result, it retains the high accuracy of two-stage detectors while significantly enhancing the detection performance and inference speed for small objects. In 2021, Srinivas et al. [13] designed a Bottleneck Attention Module based on the Transformer architecture [14], innovatively integrating convolution operations with a self-attention mechanism. The module first extracts fine-grained low-level features using multiple convolutional layers, then applies self-attention to focus on key regions within the feature maps, and finally uses downsampling to strengthen the model's semantic understanding of small objects. By modelling long-range spatial dependencies, this module effectively captures global contextual relationships, providing a new feature representation approach for small object detection in complex backgrounds. In 2022, Li et al. [15] introduced an adaptive point learning detection framework to address the geometric diversity of objects in UAV images. Their method uses a directional point feature representation to extract multi-directional geometric features and applies a spatial constraint loss function to penalise redundant points outside the oriented detection boxes. This approach preserves the geometric integrity of objects and significantly improves localisation accuracy under complex viewing angles, achieving a notable breakthrough in detecting irregularly shaped objects in oblique UAV imagery.

In 2024, Tahir et al. [16] addressed the challenges of detecting small and occluded objects by introducing PVswin-YOLOv8s, which combines YOLOv8s with the Swin Transformer. By integrating the CBAM attention mechanism and Soft-NMS, the model enhances feature extraction and occlusion handling, achieving a 4.8% increase in mAP@0.5 over YOLOv8s. To improve the extraction of small-object features and the fusion of spatial and semantic information in UAV imagery, Luo et al. [17] developed a YOLOv8n-based model incorporating Reparameterized Multi-scale Inverted Blocks, a wave-feature pyramid network, a small object detection head, and a trifocal loss function. This model, with 4.46M

parameters, reached an mAP@0.5 of 29.3%, surpassing the baseline by 3.6%. Li et al. [18] proposed SOD-YOLO to tackle uneven object distribution, cluttered backgrounds, and tiny targets. It integrates a receptive field attention module and a new neck design—BSSI-FPN—for better multi-scale feature fusion. On VisDrone2019, SOD-YOLO-s improved mAP@0.5 by 3% over YOLOv8s while cutting parameters and computation by 84.2% and 30%, respectively; SOD-YOLO-l improved accuracy by 7.7% over YOLOv8l with a 59.6% parameter reduction. Aiming to boost UAV image utility in domains like military reconnaissance and urban planning, Tan et al. [19] introduced a multi-scale detection model with adaptive feature extraction, weighting fusion, and an extra detection layer for small objects. This model achieved an mAP@0.5 of 38.5%, outperforming YOLOv5s by 2.0%.

To overcome challenges related to limited feature representation, complex backgrounds, and transmission constraints in UAV imagery, Xu et al. [20] introduced ESOD-YOLO in 2025. This lightweight detector incorporates Deformable and Global–Local Feature Guidance, Simple Cross-scale and Cross-layer Feature Fusion, and an Inverse-residual Shared Parameter module built on Partial Convolution. These additions significantly strengthen multi-scale feature fusion, delivering a strong balance between accuracy and efficiency. In a separate effort to improve the localisation of small targets, Hou et al. [21] proposed αS-YOLO—a YOLO-based architecture enhanced with a 2Filters-Global Context and Efficient Channel Attention module, a cross-convolution design, and a novel loss function named α-SIOU. Experimental results confirmed gains of 1% in mAP@0.5 and 1.1% in mAP@0.5:0.95 over leading YOLO models. Yang et al. [22] presented ADD-YOLO, built upon YOLOv8s, which leverages components like AKConv, C2f-DRAC, and CIoU-Soft-NMS to better handle background interference and extract multi-scale information effectively.

In summary, improved two-stage object detection algorithms have achieved certain advantages in terms of accuracy, but their detection speed is often less satisfactory. On the other hand, single-stage object detectors offer faster speed and stronger real-time performance, though they typically exhibit slightly lower accuracy. Therefore, this study aims to enhance the baseline YOLOv8n model to further optimise the balance between detection accuracy and speed, ultimately achieving superior overall performance.

III. METHODS

YOLOv8, released as an open-source project by Ultralytics in 2023, represents a significant update following YOLOv5. In this study, YOLOv8n is selected as the base framework instead of later versions such as YOLOv10, YOLOv11, or YOLOv12, based on the following technical considerations. First, in terms of engineering stability and deployment maturity, YOLOv8 has undergone large-scale industrial validation over the past two years. Its PyTorch implementation offers comprehensive documentation, pretrained models, and a well-established deployment toolchain. Second, YOLOv8 has relatively low computational demands during training, making it cost-effective and more accessible for experimentation. Finally, it strikes a practical balance between robustness and architectural flexibility, enabling this study to focus on core innovations rather than spending resources on compatibility issues posed by newer, less mature frameworks.

To improve detection accuracy on UAV imagery, enhancements were made to the YOLOv8n architecture. The improved model not only maintains low parameter count and fast computation speed but also significantly boosts detection accuracy. The overall architecture of the proposed algorithm is illustrated in Figure 2.

Specifically, this study introduces the following improvements and optimizations to the YOLOv8n model.

(1) Feature Map Size Adjustment. UAV images often contain a large number of small objects, and the baseline YOLOv8n model demonstrates limited accuracy in detecting such targets. To address this, we introduced a 160 $\times$ 160 resolution feature layer into the feature pyramid network, significantly enhancing the model's ability to preserve spatial details in shallow features. Compared to the smallest feature map size in the original YOLOv8 (e.g., 80 $\times$ 80), this modification enables the model to capture more pixel-level fine details (such as edges and textures of small objects), effectively mitigating feature loss caused by downsampling-an issue common in UAV imagery. In addition, we removed the 20 $\times$ 20 resolution feature layer to reduce unnecessary computation and eliminate redundant parameters.

(2) Enhanced Spatial Pyramid Pooling. The original Spatial Pyramid Pooling Fusion (SPPF) module exhibits limited capability in extracting features from multi-scale targets. To address this limitation, we incorporated the Large Separable Kernel Attention (LSKA) module into the SPPF. The large-kernel separable convolutional attention mechanism significantly enhances the multi-scale feature extraction ability of the SPPF module, thereby improving the model's capacity to accurately detect small objects in UAV imagery.

(3) Soft-NMS and MPDIoU. Soft-NMS is an improved version of the traditional Non-Maximum Suppression (NMS). Unlike standard NMS, it attenuates the confidence scores of overlapping bounding boxes rather than suppressing them entirely, thus reducing missed detections, particularly in dense target scenarios. MPDIoU [23], in contrast to traditional Complete-IoU (CIoU) [24], calculates IoU based on the minimum point distance of horizontal rectangles, allowing it to simultaneously consider the overlap area, the distance between central points, and deviations in width and height. The combination of Soft-NMS and MPDIoU effectively enhances the detection of overlapping and occluded objects in UAV imagery.

In summary, the improvements made to the YOLOv8n model have enhanced its ability to detect small objects while also addressing the challenge of detecting overlapping targets. The integration of these enhancements has

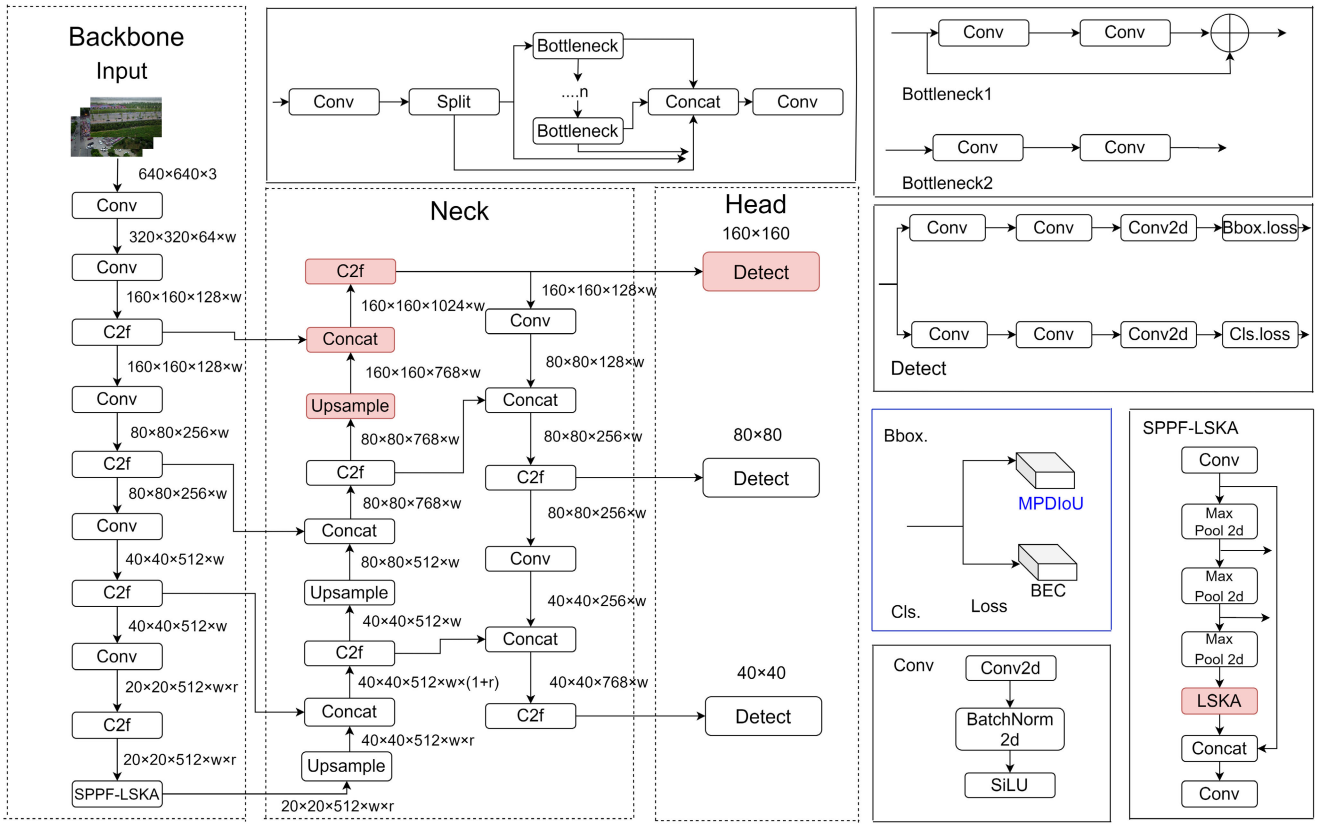


FIGURE 2. The network architecture of the improved YOLOv8n model.

collectively contributed to a significant improvement in detection accuracy for UAV imagery.

A. MULTI-SCALE NETWORK RECONSTRUCTION

In the VisDrone2019 dataset, small objects constitute the majority. Extensive experimental results indicate that the baseline YOLOv8n model exhibits certain limitations when applied to this dataset:

(1)Insufficient shallow feature representation. Deep convolutional downsampling during forward propagation through the backbone significantly attenuates critical features, such as the edges and textures of small objects.

(2) Limited adaptability of multi-scale detection heads. The baseline YOLOv8 employs feature map sizes of 80 × 80, 40 × 40, and 20 × 20, corresponding to targets larger than 8 × 8, 16 × 16, and 32 × 32 pixels, respectively. However, annotated objects in UAV images are often smaller than 32 × 32 pixels. This mismatch between feature map resolution and target size results in suboptimal detection accuracy.

To address the challenges of target scale variation and small object feature loss in UAV image detection, this study applies a multi-scale feature enhancement approach, with a particular focus on optimizing the Neck structure in YOLOv8n and extending the high-resolution detection heads. As shown in Figure 3, the improved YOLOv8n model includes the following technical modifications:

(1) Cross-layer feature fusion reconstruction in the Neck. Based on the dual-pyramid FPN-PAN structure of the baseline YOLOv8n, this study enhances the reuse of shallow feature maps. High-resolution features from the 160×160 output of the backbone are extracted and integrated into the FPN network via lightweight cross-layer connections (including 1×1 convolutions for channel compression and upsampling). This design better preserves the fine-grained edges and texture details of small objects, while mitigating the loss of semantic information caused by deep downsampling.

(2) Small object detection head design. In addition to the existing 8×8, 16×16, and 32×32 detection heads, a 4×4 pixel-level detection head is introduced to enhance the detection capability for small objects. At the same time, the 32×32 detection head is removed to avoid miscalculations in image sizing and reduce model computational complexity and parameter overhead.

In summary, by adding a 160×160 feature map and a 4×4 detection head, the model significantly improves its ability to extract semantic information from small targets and achieves better detection accuracy for small objects.

B. THE SPPF-LSKA MODULE

To address the challenges in UAV images—specifically, the high proportion of small objects, blurry features, and susceptibility to interference from complex backgrounds—we propose a targeted enhancement to the SPPF module

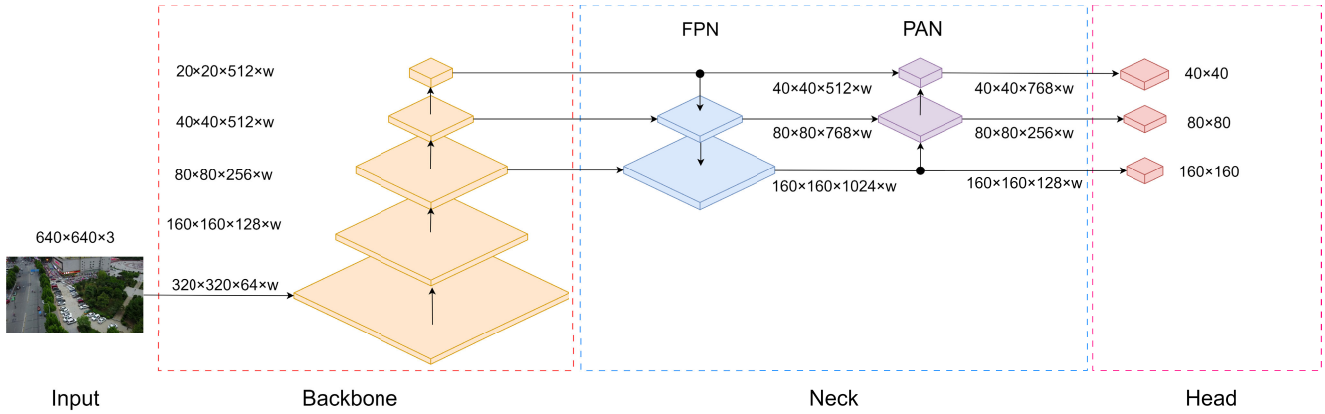


FIGURE 3. Improved multi-scale network structure.

in YOLOv8n. By embedding the Large Separable Kernel Attention (LSKA) mechanism, we significantly boost the model's multi-scale feature fusion capability and semantic representation accuracy for small objects, all while maintaining a lightweight architecture.

The original SPPF module uses three 5×5 max pooling layers to extract multi-scale contextual features. While this structure captures local multi-granular information, it exhibits limited receptive field coverage and insufficient long-range dependency modeling, which are crucial for detecting small objects commonly found in UAV scenarios.

(1) Dilution of Small Object Features. During the downsampling process, small objects occupy a low pixel proportion, making them prone to feature blurring due to pooling operations.

(2) Loss of Contextual Association. Relying solely on fixed-scale pooling limits the ability to model spatial relationships across regions—such as the semantic association between small objects and their surrounding environment.

The improved module builds upon the original multi-scale pooling in SPPF by adding a separable large-kernel convolution attention branch. The detailed structure is illustrated in Figure 4.

The SPPF-LSKA module has the following characteristics:

(1) Multi-scale feature extraction. It retains the 3-layer 5×5 max pooling structure of the original SPPF to generate feature maps at different scales, which are then concatenated to form the initial multi-scale feature representation.

(2) Separable large-kernel convolution. The traditional large kernel K×K convolution is replaced with a sequential operation of a 1×K row convolution followed by a K×1 column convolution, significantly reducing computational complexity.

(3) Attention-based weight allocation. After the separable convolution, a lightweight channel attention module is introduced. It uses global average pooling and an adaptive gating mechanism to assign weights to different channels of the large-kernel features, enhancing the response of channels sensitive to small-object details (such as edges and high-frequency textures).

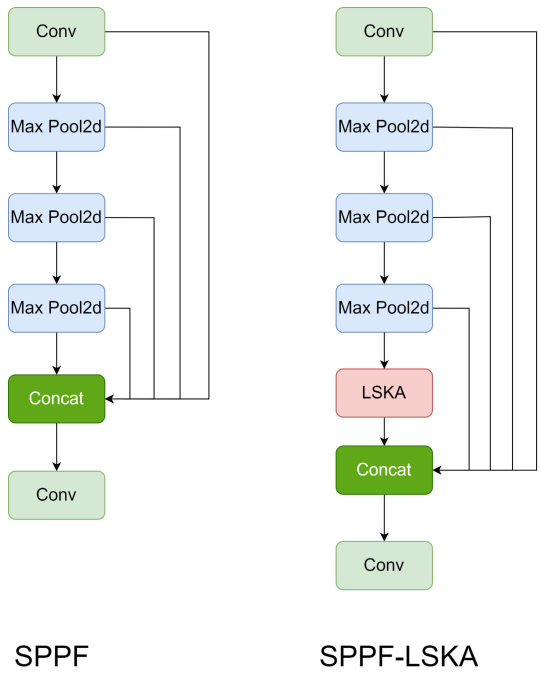


FIGURE 4. SPPF and SPPF-LSKA modules.

From a mathematical perspective, the feature transformation in the SPPF-LSKA module can be represented by Equation (1).

$$F_{LSKA} = \text{Attn}\left(\text{Conv}_{1\times K}\left(\text{Conv}_{K\times 1}(F_{Pool})\right)\right). \quad (1)$$

where $\text{Conv}_{1\times K}$ and $\text{Conv}_{K\times 1}$ represent the separable convolutions in the row and column directions, respectively, and denotes the channel attention function.

The SPPF-LSKA module is designed to expand the effective receptive field (ERF) efficiently, crucial for small-object detection in UAV imagery. By using large separable kernel attention (LSKA), it enlarges receptive fields through cascaded 1×K and K×1 convolutions without heavy computational costs. This captures broader contextual information, preserving fine-grained details of small objects often lost in downsampling. Additionally, the channel-wise attention

enables adaptive feature fusion, emphasizing informative features while suppressing irrelevant background.

Overall, SPPF-LSKA combines efficient ERF expansion, shape-aware attention, and adaptive multi-scale feature integration to improve small-object recognition.

C. SOFT-NMS ALGORITHM

In UAV object detection, densely distributed targets often lead to highly overlapping bounding boxes. Traditional Non-Maximum Suppression (NMS) removes low-score candidate boxes by setting a fixed Intersection over Union (IoU) threshold, which can result in missed detections due to excessive suppression—particularly problematic in UAV-view scenes such as clusters of vehicles or crowds. To address this, we introduce the Soft-NMS algorithm as a replacement for conventional NMS. Instead of hard filtering, it dynamically adjusts the confidence scores of overlapping boxes, effectively preserving potential target features and significantly improving detection completeness in dense scenes.

Soft-NMS softens the suppression strategy and restructures the candidate box selection logic: for candidate boxes that overlap with the highest-score box, they are no longer directly removed. Instead, their confidence scores are gradually decayed based on the degree of overlap. Specifically, two types of decay functions are employed:

(1) Linear decay, with the formula as follows:

$$S_i = \begin{cases} S_i, & \text{if IoU}(M, b_i) < N_t \\ S_i \cdot (1 - \text{IoU}(M, b_i)), & \text{if IoU}(M, b_i) \geq N_t \end{cases} \tag{2}$$

(2) Gaussian decay, with the formula as follows:

$$S_i = S_i \cdot \exp\left(-\frac{\text{IoU}(M, b_i)^2}{\sigma}\right), \quad \forall b_i \neq M \tag{3}$$

where M is the current highest scoring box, b_i is an overlapping box, S_i is the confidence score, N_t is the IoU threshold, and σ is a hyperparameter controlling the decay rate. Compared with the "black-or-white" selection strategy of traditional NMS, Soft-NMS allows overlapping boxes to continue participating in subsequent filtering with decayed scores, thereby avoiding missed detections caused by sensitivity to IoU thresholds. This is particularly suitable for complex scenarios in VisDrone2019 imagery where object boundaries are ambiguous, scales vary, and occlusion is dense.

In VisDrone2019 imagery, targets often exhibit irregular overlaps due to oblique angles and resolution differences (e.g., dense traffic on highways, pedestrians among urban building clusters). The rigid cutoff of traditional NMS easily eliminates valid detection boxes that are highly overlapping with the main box but actually belong to different instances.

Soft-NMS mitigates this by gradually decaying the confidence scores of overlapping boxes rather than removing them outright, preserving potential target information. This approach improves detection completeness in crowded or overlapping scenes, maintaining sensitivity to fine-grained object boundaries and enhancing robustness in complex UAV imagery. By retaining gradient information in overlapping regions, Soft-NMS enables more accurate distinction between adjacent or partially occluded objects.

D. MPDIOU LOSS FUNCTION

YOLOv8 adopts a multi-task learning approach in its loss function design, comprising three key components: coordinate loss, confidence loss, and classification loss. Coordinate loss evaluates the accuracy of the bounding boxes, confidence loss reflects the certainty of the model, and classification loss measures classification accuracy. These three components are calculated individually and then summed, as shown in the equation below:

$$\text{Loss} = L_{\text{IoU}} + L_{\text{Obj}} + L_{\text{Cls}} \tag{4}$$

The baseline YOLOv8 model employs CIoU for bounding box regression. Unlike standard IoU, CIoU incorporates not only the overlap between predicted and ground truth boxes but also introduces a penalty term to improve alignment in aspect ratios. As defined in Equations (5) – (7), $\rho^2(\mathbf{e}, \mathbf{e}^{\text{gt}})$ measures the squared Euclidean distance between the centers of the two boxes, while c denotes the diagonal length of the smallest enclosing box that covers both. In this context, w and h denote the predicted box's width and height, while w^{gt} and h^{gt} represent those of the ground truth. The penalty term V is introduced to account for shape and orientation differences, aiming to refine localization accuracy. Although CIoU provides a more comprehensive evaluation by considering overlap, center distance, and aspect ratio, its V component only reflects aspect ratio deviation rather than true width and height differences influenced by confidence scores, which can sometimes limit optimization efficiency.

$$L_{\text{CIoU}} = 1 - \text{IoU} + \frac{\rho^2(\mathbf{e}, \mathbf{e}^{\text{gt}})}{c^2} + \alpha V \tag{5}$$

$$V = \frac{4}{\pi^2}\left(\arctan\frac{w^{\text{gt}}}{h^{\text{gt}}} - \arctan\frac{w}{h}\right)^2 \tag{6}$$

$$\text{Loss} = L_{\text{CIoU}} + L_{\text{Obj}} + L_{\text{Cls}} \tag{7}$$

In complex recognition scenarios, objects often appear with significant overlap. MPDIoU offers a streamlined approach to evaluating bounding box similarity, making it particularly effective for overlapping object regression by accelerating convergence and enhancing regression precision. As a result, this study adopts MPDIoU as the loss function for bounding box optimization. MPDIoU is a similarity measure that integrates overlap area, center-point distance, and discrepancies in width and height. Unlike traditional metrics, it excels in differentiating boxes that share the same aspect ratio but vary in scale or position. By directly measuring distances between critical points of predicted and ground truth boxes, MPDIoU delivers a more accurate and discriminative loss signal. The formula for MPDIoU is

presented as follows:

$$d_1^2 = (x_1^p - x_1^t)^2 + (y_1^p - y_1^t)^2 \tag{8}$$

$$d_2^2 = (x_2^p - x_2^t)^2 + (y_2^p - y_2^t)^2 \tag{9}$$

$$\text{MPDIoU} = \text{IoU} - \frac{d_1^2}{h^2 + w^2} - \frac{d_2^2}{h^2 + w^2} \tag{10}$$

In Equations (8) – (10), variables d_1 and d_2 represent the Euclidean distances between the top-left and bottom-right corners of the two boxes (as shown in Figure 10). Variables w and h represent the input image dimensions. Coordinates (x_1^t, y_1^t), (x_1^p, y_1^p), (x_2^t, y_2^t) and (x_2^p, y_2^p) represent the top-left and bottom-right corners of the ground truth and predicted boxes, respectively, as shown in Figure 10. As can be seen, all factors required for the current loss function calculations can be derived from coordinate distances between the prediction and ground truth boxes, including individual areas, center point distances, and size differences. This implies that MPDIoU is both highly generalizable and computationally simple.

When MPDIoU is used as the loss function, its calculation principle is expressed as follows. Meanwhile, the structural schematic is shown in Figure 5.

$$L_{\text{MPDIoU}} = 1 - \text{MPDIoU} \tag{11}$$

The MPDIoU loss function refines bounding box regression by considering overlap, center-point distances, and width/height differences simultaneously. Compared with CIoU, MPDIoU provides a more direct and discriminative measurement of box similarity, particularly effective in scenarios with overlapping or closely positioned objects. Its formulation accelerates convergence and enhances localization accuracy, ensuring that both position and scale differences are properly penalized. Additionally, MPDIoU is computationally efficient and generalizable, relying solely on coordinate distances, making it highly suitable for dense UAV object detection tasks.

E. INTEGRATED MODULE OVERVIEW

To improve small-object detection in UAV imagery, this study introduces several enhancements based on YOLOv8. The small-object detection head was replaced to increase the model's sensitivity to fine-grained target features, thereby boosting detection accuracy. The SPPF-LSKA module expands the receptive field through large separable kernel attention and incorporates channel-wise attention for adaptive multi-scale feature fusion, strengthening the model's response to object edges and textures. Soft-NMS gradually reduces the confidence scores of overlapping boxes instead of discarding them outright, preserving information in dense or occluded scenes and improving detection completeness. The MPDIoU loss function simultaneously accounts for overlap, center-point distance, and width-height differences in bounding box regression, enhancing localization precision and accelerating convergence. Collectively, these improvements work together to enhance accuracy, robustness, and completeness in UAV small-object detection while maintaining computational efficiency.

IV. EXPERIMENT

A. DATASETS

The VisDrone2019 dataset, constructed by the Machine Learning and Data Mining Lab at Tianjin University, is a significant benchmark in the field of UAV-based aerial imagery. The dataset includes annotations for over 2.6 million common objects, providing both point coordinates and bounding boxes for various small target categories such as pedestrians and distant objects.

The data was captured using cameras mounted on different types of drones, across 14 diverse urban and rural locations in China. It covers a wide range of environmental conditions, including sunny, cloudy, and rainy weather, as well as varying lighting scenarios such as strong daylight and low-light nighttime scenes. This diversity reflects the complex and realistic settings of everyday life.

The dataset is divided into 6,471 images for training, 548 for validation, and 1,610 for testing. The original annotations contain 12 object categories, but two of them "other" and "ignored regions" do not correspond to actual detection targets and are removed before training. After preprocessing, 10 object categories remain: pedestrian, people, bicycle, car, van, truck, tricycle, awning-tricycle, bus, and motor. The distribution of instances across these object categories is shown in Figure 6.

B. EXPERIMENTAL ENVIRONMENT

All experiments were carried out on a Linux-based server equipped with an NVIDIA GeForce RTX 3090 GPU. The training leveraged CUDA 11.7 and was implemented using the PyTorch 1.13.1 framework. Input images were resized to 640×640 pixels, with a batch size of 16. The SGD optimizer was used with a momentum of 0.937 across 200 training epochs. An initial learning rate of 0.01 was applied, incorporating Cosine Annealing and Linear Decay strategies for dynamic adjustment during training. as shown in Table 1.

TABLE 1. Experimental configuration.

Component	Name/Value
Operating system	Linux
GPU	NVIDIA GeForce RTX3090
CUDA	11.7
PyTorch	1.13.1
Input image size	640 × 640
Training batch size	16
Optimizer momentum	0.937
Epochs	200
Learning rate	0.0092
Learning rate decay strategy	Cosine Annealing and Linear Decay

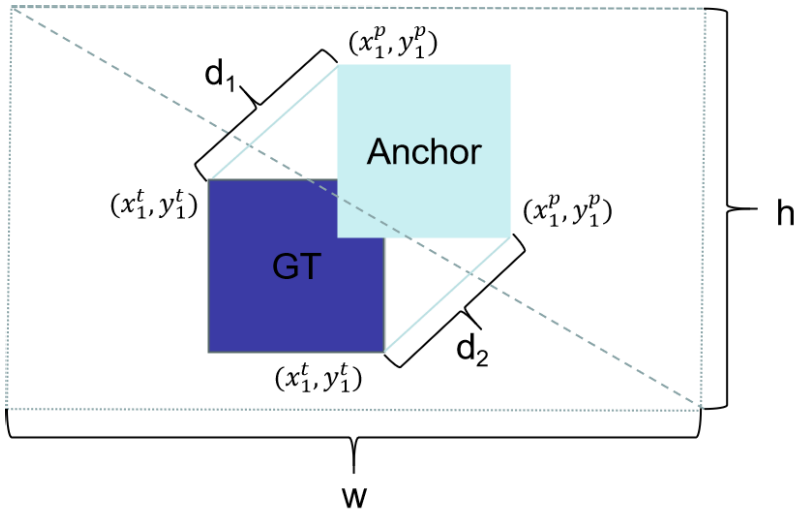


FIGURE 5. MPDIoU structural diagram.

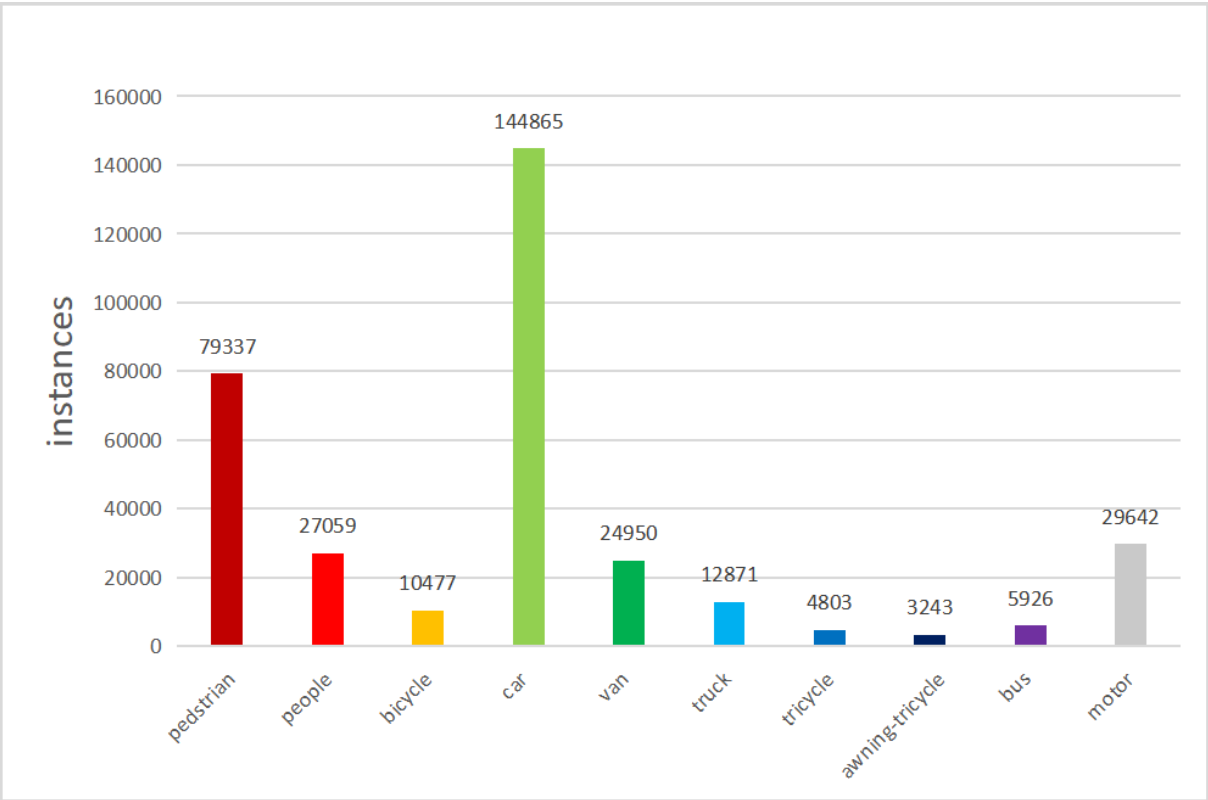


FIGURE 6. Instance distribution of each category in the VisDrone2019 dataset.

C. EVALUATION METRICS

The performance evaluation metrics commonly used to assess the effectiveness of object detection algorithms include: Recall (R), Precision (P), mean Average Precision (mAP) across all classes, number of parameters (Params), computational complexity (GFLOPs), model size, and detection speed (FPS). Among these, mAP is considered the most critical metric in object detection tasks.

Precision measures the proportion of true positive detections among all predicted positives, indicating the accuracy of the model's positive predictions. Recall, on the other hand, reflects the proportion of true positives identified out of all

actual positives, assessing the model's ability to capture all relevant targets. Their respective formulas are given below:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{12}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{13}$$

True Positive (TP) denotes the count of real positive instances accurately detected by the algorithm. False Negative (FN) represents actual positive instances that the model fails to detect, misclassifying them as negatives. False Positive (FP) refers to negative instances that are mistakenly identified as positives—indicating false alarms.

The Precision-Recall (P-R) Curve visually depicts the trade-off between precision and recall. The Average Precision (AP) is computed as the area under this curve for each class, providing a holistic measure of detection performance across varying recall thresholds. A higher AP score signifies that the model consistently achieves high precision over a broad recall range. The AP is calculated using the following formula.

$$AP = \int_0^1 p(r)\, dr \tag{14}$$

$$\text{mAP} = \frac{1}{C} \sum_{c=1}^{C} \text{AP}_c \tag{15}$$

Mean Average Precision (mAP) is a key evaluation metric in object detection. It is obtained by averaging the Average Precision (AP) scores across all object categories in the dataset. This metric offers an overall assessment of the model's detection performance across various classes. The detailed calculation is shown in Equation (15).

Frames Per Second (FPS) is commonly used as an efficiency metric for object detection algorithms. It reflects the detection speed of the algorithm under a specific hardware configuration. Generally, a higher FPS indicates faster detection performance.

Giga Floating Point Operations per Second (GFLOPs) is an important metric for measuring a model's computational complexity. For YOLOv8, a lower GFLOPs value usually indicates a more efficient model that can process images faster under the same hardware conditions.

In many practical applications, model size is an important consideration, especially on resource-constrained devices. For example, on mobile devices or embedded systems, a smaller model is easier to deploy and runs more efficiently.

D. EXPERIMENTAL RESULTS ANALYSIS

1) EFFECTIVENESS ANALYSIS OF EACH METHOD

(1) Effectiveness Analysis of Multi-Scale Network Reconstruction.

UAV imagery typically contains a high proportion of small objects, and the baseline YOLOv8n model struggles to effectively extract their features, often resulting in false negatives and misclassifications. To mitigate this issue, this study introduces a 160×160 resolution feature map along with a 4×4 pixel-level detection head to strengthen small-object feature extraction. The effectiveness of this enhancement is evaluated on the VisDrone2019 dataset, with the detection performance summarized in Table 2.

As shown in Table 2, after reconstructing the multi-scale network, the model's mAP@0.5 and mAP@0.5:0.95 increased by 2.2% and 1.6% respectively, while the number of parameters decreased by 33%. However, due to the addition of the 160×160 feature map, the computational complexity of the model increased, resulting in an additional 3.7 GFLOPs compared to the baseline YOLOv8n. Overall, our method shows greater advantages in detecting small objects in UAV images, as mAP@0.5 is the most critical evaluation metric.

(2) Effectiveness Analysis of the SPPF-LSKA Module.

In UAV images, small targets account for a high proportion, often appearing with blurred features and are easily affected by complex backgrounds. Although the baseline YOLOv8n's SPPF module can extract multi-scale features, it lacks sufficient receptive field and long-range dependency modeling for small targets. By integrating the LSKA (Large Separable Kernel Attention), the module decomposes large-kernel convolutions to reduce parameters while expanding the receptive field, and uses attention mechanisms to enhance feature representation. We conducted a comparative analysis of the SPPF and SPPF-LSKA modules under the same conditions, and the detection results are shown in Table 3.

By observing Table 3, it can be concluded that compared to the original SPPF, the SPPF-LSKA module improves mAP@0.5 and mAP@0.5:0.95 by 0.3% and 0.2%, respectively. Meanwhile, the number of parameters increases by 0.2M and the computational complexity rises by 0.2G. Although the SPPF-LSKA module does not significantly enhance the model's detection performance, the increase in computational complexity is minimal. Therefore, this module is retained for further integration and analysis to determine its compatibility with the overall algorithm.

(3) Effectiveness Analysis of the Soft-NMS Algorithm.

In UAV image object detection tasks, densely distributed targets often lead to significant overlap among detection bounding boxes. Traditional Non-Maximum Suppression (NMS) directly filters out low-confidence candidate boxes based on a fixed IoU threshold, which can result in missed detections due to excessive suppression. This issue is particularly prominent in dense scenes commonly observed from aerial perspectives, such as clusters of vehicles or crowds of people. To address this, we adopt the Soft-NMS algorithm as a replacement for traditional NMS. Instead of hard removal, Soft-NMS dynamically adjusts the confidence scores of overlapping candidate boxes through a decaying function, effectively preserving potential target features and significantly improving detection completeness in dense scenarios. We conducted comparative experiments between the two methods, and the results are shown in Table 4.

The results in Table 4 demonstrate that compared to traditional NMS, the Soft-NMS algorithm significantly improves detection accuracy, with mAP@0.5 and

TABLE 2. Analysis of the effectiveness of multi-scale network reconstruction.

Methods	mAP@0.5(%)	mAP@0.5:0.95(%)	Params(M)	GFLOPs
YOLOv8n	34.5	20.1	3.0	**8.1**
Ours	**36.7**	**21.7**	**2.0**	11.8

Bold font indicates the best values.

TABLE 3. Effectiveness analysis of the SPPF-LSKA module.

Methods	mAP@0.5(%)	mAP@0.5:0.95(%)	Params(M)	GFLOPs
YOLOv8n	34.5	20.1	**3.0**	**8.1**
Ours	**34.8**	**20.3**	3.2	8.3

Bold font indicates the best values.

mAP@0.5:0.95 increasing by 5.9% and 6%, respectively. There is no change in the number of parameters or computational complexity, indicating that Soft-NMS does not introduce complex computational procedures. Instead, it simply applies a different strategy for handling overlapping regions. Therefore, this study adopts the Soft-NMS algorithm as the post-processing step in the improved model.

(4) Effectiveness Analysis of MPDIoU.

In UAV image object detection, the accuracy of bounding box regression directly determines the effectiveness of detecting small and densely overlapping targets. Although the traditional CIoU loss incorporates Intersection over Union (IoU), center point distance, and aspect ratio constraints, it has significant limitations when applied to UAV-specific scenarios involving oblique-view rotating objects, extreme scale variations, and complex occlusions. CIoU's geometric constraint model struggles to effectively capture features such as target rotation angles, dynamic changes in width-to-height ratios, and overlapping regions caused by occlusions. This results in significant issues with small object localization errors and boundary confusion among dense targets. To address this, we adopt the MPDIoU loss function as a replacement for CIoU. MPDIoU introduces multi-perspective geometric constraint terms based on CIoU, integrating target rotation angles, dynamic scale weighting, and occlusion-aware mechanisms. This constructs a bounding box regression model tailored to the oblique imaging characteristics of UAVs, effectively enhancing localization accuracy and robustness under complex geometric conditions. The comparison results of the two IoU loss functions are shown in Table 5.

As shown in Table 5, MPDIoU improves detection performance over the baseline CIoU, with increases of 0.2% in mAP@0.5 and 0.1% in mAP@0.5:0.95. Importantly, these gains come without additional parameters or computational overhead, making MPDIoU a more effective choice for UAV-based object detection. Accordingly, this study adopts MPDIoU as the loss function in the improved model.

2) ABLATION EXPERIMENT

To assess the individual contributions and overall compatibility of the proposed enhancements, ablation experiments were performed on a custom dataset under consistent conditions. These tests aimed to evaluate how each modification influences model performance. Table 6 summarises the results, where components A, B, C, and D correspond to multi-scale network reconstruction, the SPPF-LSKA module, the Soft-NMS algorithm, and the MPDIoU loss function, respectively.

From the experimental data in Table 6, it is evident that the improved model demonstrates significantly better detection accuracy on the VisDrone2019 dataset compared to the baseline YOLOv8n. However, in terms of computational complexity, the improved model is lower than the baseline YOLOv8n, indicating that the proposed enhancements have introduced additional algorithmic complexity to some extent. A detailed analysis of each individual improvement strategy shows that all proposed modifications contributed to varying degrees of improvement in detection accuracy. Specifically, reconstructing the multi-scale detection network and replacing the SPPF module with SPPF-LSKA both led to accuracy improvements but also increased the model's computational cost. Overall, when all the proposed modules were integrated, the improved YOLOv8n achieved optimal detection accuracy, clearly demonstrating that the enhanced components are highly compatible and can effectively work together to boost model performance.

In summary, although there is a slight trade-off in inference speed, the proposed algorithm delivers substantial performance gains for UAV image detection tasks, offering a more practical and feasible solution for real-world application.

The data presented in Figure 7 illustrates the detection accuracy for various annotated object categories. The improved YOLOv8n model demonstrates significant accuracy gains in detecting pedestrians, people, bicycles, and motorcycles. However, for cars, buses, and vans, the improvements are minimal or even slightly decreased. This indicates that while the proposed improvements enhance small object detection accuracy, the removal of the large-object detection head leads to a reduction in accuracy for larger targets. Table 7 below provides a clearer view of the average precision differences before and after the improvements for each category.

TABLE 4. Effectiveness analysis of the soft-NMS algorithm.

Methods	mAP@0.5(%)	mAP@0.5:0.95(%)	Params(M)	GFLOPs
YOLOv8n	34.5	20.1	3.0	8.1
Ours	**40.4**	**26.1**	3.0	8.1

Bold font indicates the best values.

TABLE 5. Effectiveness analysis of the MPDIoU algorithm.

Methods	mAP@0.5(%)	mAP@0.5:0.95(%)	Params(M)	GFLOPs
YOLOv8n	34.5	20.1	3.0	8.1
Ours	**34.7**	**20.2**	3.0	8.1

Bold font indicates the best values.

TABLE 6. Ablation Experiment Results.

Model	A	B	C	D	mAP@0.5 (%)	mAP@0.5:0.95 (%)	GFLOPs	Params (M)	FPS	Size (MB)
YOLOv8s					34.5	20.1	8.1	3.0	**51.82**	6.2
	✓				36.7	21.7	11.8	**2.0**	**–**	**4.5**
		✓			34.8	20.3	8.3	3.2	–	6.8
			✓		40.4	26.1	8.1	3.0	–	6.2
				✓	34.7	20.2	8.1	3.0	–	6.2
	✓	✓	✓		42.0	26.8	12.0	2.3	–	5.0
Ours	✓	✓	✓	✓	**42.1**	**27.7**	**12.0**	2.3	49.12	5.0

TABLE 7. Comparison of Class-wise Precision between two algorithms.

Category	YOLOv8n(%)	Ours(%)
pedestrian	36.2	45.4
people	29.1	40.9
bicycle	8.2	20.8
car	76.1	74.3
van	39.6	45.4
truck	31.1	39.7
tricycle	24.6	32.9
Awning-tricycle	13.0	19.8
bus	49.1	56.0
motor	37.7	45.6

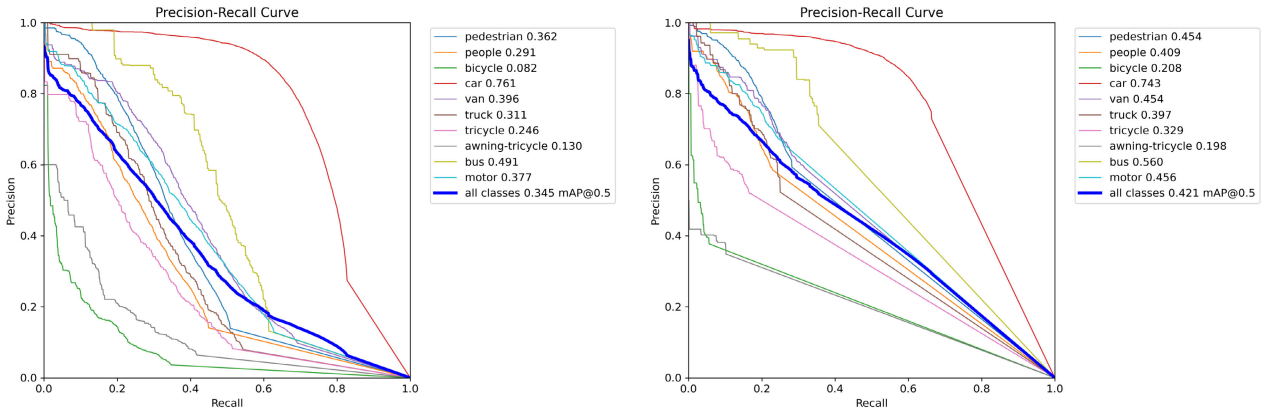
Figure 8 shows the line chart of mAP@0.5 before and after the improvement. The green line represents the improved YOLOv8n model, while the blue line represents the baseline YOLOv8n. It can be observed that, compared to the baseline, the improved model not only converges faster but also achieves significantly higher accuracy. In contrast, although the baseline YOLOv8n shows a rapid increase in accuracy during the early training stages, its performance plateaus after 50 epochs and converges relatively slowly, requiring more training time. Overall, the proposed algorithm demonstrates a significant advantage in UAV target detection.

3) VISUALIZATION ANALYSIS

We compared the detection results of the improved YOLOv8n model with those of the baseline YOLOv8n model. Three images were selected from the VisDrone2019 dataset, representing different scenes, lighting conditions, overlapping objects, and small targets. Key detection areas are highlighted with pink boxes, and different colors represent different categories to illustrate the detection performance.

Figure 9 shows a parking lot scene captured by a UAV during daytime. The image is taken from an oblique angle looking into the distance, where many of the targets appear very small—commonly referred to as small objects. Comparing the results of the baseline YOLOv8n and the improved model, we observe that the baseline model misdetects an object in the lower-left corner and misses a detection in the upper-right area. In contrast, our improved model correctly detects all relevant targets in both areas, demonstrating superior performance in detecting small objects in UAV imagery.

Figure 10 shows a street scene captured by a drone at dusk, presented from an overhead perspective. In the image, a large crowd of people is gathered closely together, causing significant occlusion and overlap, and these are also considered small targets. When comparing the baseline YOLOv8n and the improved model, it can be observed that YOLOv8n produces false detections in the upper-left corner, as well as a large number of false positives in the center and lower-right regions. In contrast, the improved model avoids false detections in the upper-left area, but still shares similar errors with YOLOv8n in the center, and produces a few false positives in the lower-right. This indicates that while the improved model performs better than YOLOv8n in handling overlapping and occlusion scenarios, there is still room for improvement in accurately detecting overlapping and occluded targets.



(a) YOLOv8n (b) Ours

FIGURE 7. P-R curves of YOLOv8n before and after improvement.

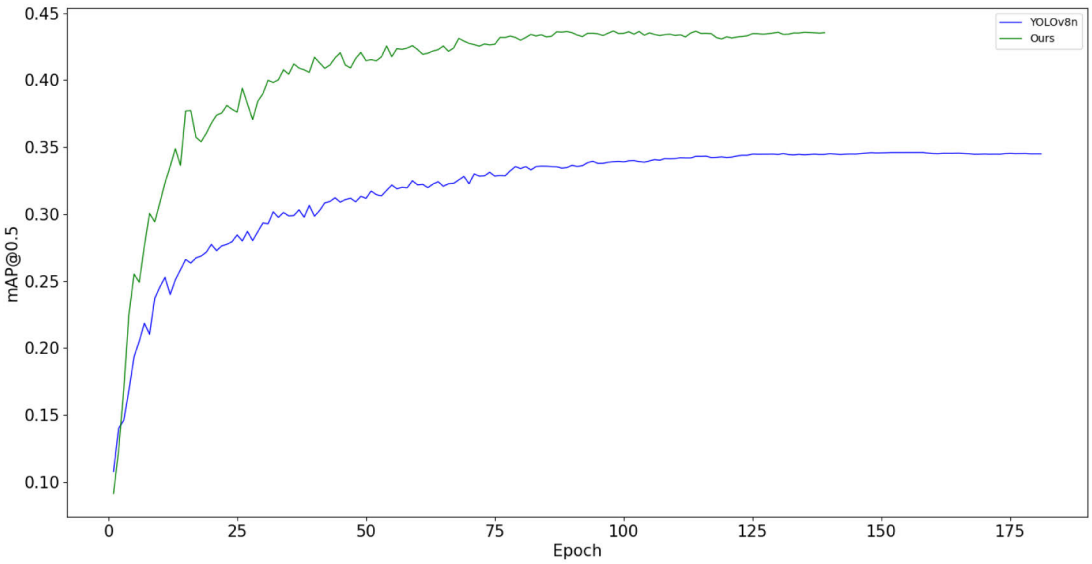


FIGURE 8. Line chart of mAP@0.5 for YOLOv8n before and after improvement.

(a) Original Image (b) YOLOv8n (c) Ours

FIGURE 9. Small Objects at an oblique angle during daytime.

Figure 11 shows a plaza scene captured by a drone at night. This image is taken from an oblique angle, and many people are gathered in the plaza. The image is pixelated and blurry, which presents challenges for object detection. Comparing YOLOv8n with the improved model, we can see that YOLOv8n makes multiple false detections in the lower-left and central areas, misidentifying shadows as pedestrians. In the upper-right area, it incorrectly classifies a streetlamp as a pedestrian. In contrast, our improved model makes only one false detection in the lower-left, and performs

accurately in the central and upper-right areas. This indicates that our model outperforms YOLOv8n when dealing with blurred or low-resolution images.

E. COMPARISON BETWEEN DIFFERENT ALGORITHMS

1) COMPARISON WITH TRADITIONAL OBJECT DETECTION ALGORITHMS

To further validate the superiority of the proposed algorithm, we conducted extensive experiments on the same dataset, VisDrone2019, and compared it with traditional object detection models commonly used in UAV applications, including CenterNet [25], RetinaNet [26], Cascade R-CNN [27] and Faster R-CNN [28]. The experimental results are presented in Table 8.

The enhanced YOLOv8n model demonstrates superior performance compared to other detection algorithms. Faster R-CNN, which uses a Region Proposal Network (RPN) to generate candidate regions, often underperforms on small object detection due to limited ability to accurately localise tiny targets, leading to missed detections or localization errors. Cascade R-CNN improves accuracy through multi-stage refinement, but its computational cost rises significantly with each stage, limiting its real-time applicability—especially when processing small objects that require finer granularity.

RetinaNet addresses class imbalance using focal loss, but its single-stage framework struggles to capture the subtle features of small objects, which are often diluted by a large volume of background data. Similarly, CenterNet, which detects objects via keypoints, has difficulty identifying the center points of small targets, as these are easily affected by noise, reducing localization accuracy.

2) COMPARISON WITH IMPROVED YOLO SERIES OBJECT DETECTION ALGORITHMS

To more comprehensively evaluate the performance of the improved YOLOv8n model, we selected several of the most advanced YOLO-based object detection algorithms for comparison. These include small-scale models such as LE-YOLO [29], SOD-YOLO-n [18], ESOD-YOLO [20] YOLO11n [32] and PS-YOLO-n [33], as well as medium-scale models like YOLOv5-pp [30], UAV-YOLO [31] and TOOD [34].

As shown in Table 9, the improved YOLOv8n model delivers better detection performance than other enhanced variants. It surpasses the lightweight LE-YOLO by 2.8% in mAP@0.5 and 5% in mAP@0.5:0.95, while reducing computational complexity by 1.1 GFLOPs and adding only 0.2M parameters. Compared with SOD-YOLO-n, it achieves gains of 9.1% in mAP@0.5 and 8.4% in mAP@0.5:0.95, with a moderate increase of 1.7M parameters and 4.2 GFLOPs. Comparable improvements are also observed over ESOD-YOLO and YOLO11n. PS-YOLO-n is an improved version based on YOLOv11n. The number of parameters is reduced by 46%, and the computational complexity is decreased by 1.2 GFLOPs, indicating that PS-YOLO-n achieves better lightweight performance. However, the improvement in detection accuracy is marginal, with an increase of only 0.1%, suggesting a limited gain in precision.

When compared with mid-sized models such as YOLOv5-pp, although the mAP@0.5 improvement is relatively small (0.4%), the proposed model requires only 21.9% of its parameter count. Furthermore, it surpasses USV-YOLO by 3.6% in mAP@0.5, while reducing parameters and computation by 5.2M and 5.7 GFLOPs, respectively. However, TOOD achieves relatively high small object detection accuracy, slightly lower than our model, but it has a much larger number of parameters and higher computational complexity, resulting in poor real-time detection performance.

Overall, the improved YOLOv8n model not only achieves the highest detection accuracy among the evaluated models but also strikes an excellent balance between accuracy and computational efficiency.

F. GENERALIZATION TEST OF THE PROPOSED ALGORITHM

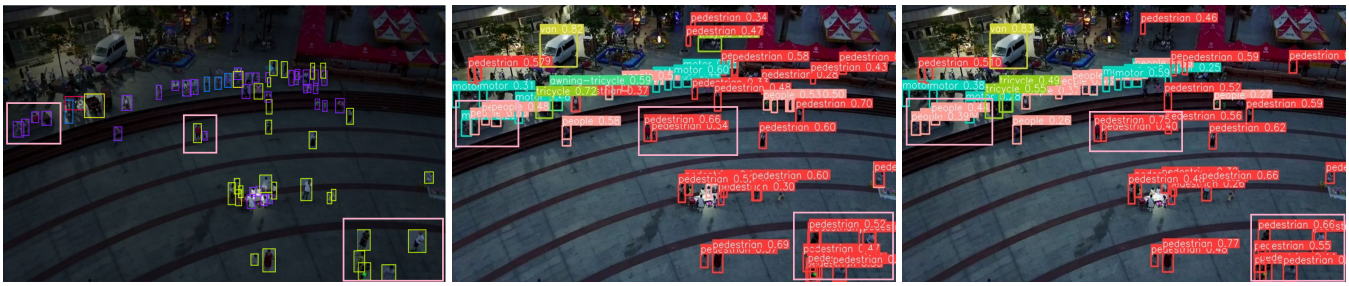
To evaluate the generalization capability of the proposed algorithm across different datasets, we conducted additional experiments on the publicly available RSOD dataset. RSOD is a small-scale aerial image dataset consisting of 976 images, which were split into training, validation, and test subsets with a ratio of 8:1:1. Under the same experimental settings, we compared our method with several representative baselines, including YOLOv8n, YOLOv10n, Deformable DETR, and RT-DETR-r18. The experimental results are presented in Table 10.

As shown in Table 10, the improved YOLOv8n model achieved the best overall performance, with a 3.4% increase in mAP@0.5, while also having the fewest parameters and the smallest model size. Both YOLOv8n and YOLOv10n demonstrate high detection accuracy with the lowest parameter count and computational complexity, indicating that these models are relatively mature for small object detection tasks. Although Deformable DETR [1007] and RT-DETR-r18 also achieved relatively high detection accuracy, their computational cost and model size are significantly higher compared to the other methods. In summary, the proposed algorithm exhibits excellent performance on the RSOD dataset as well, demonstrating its strong generalization capability.

V. DISCUSSION

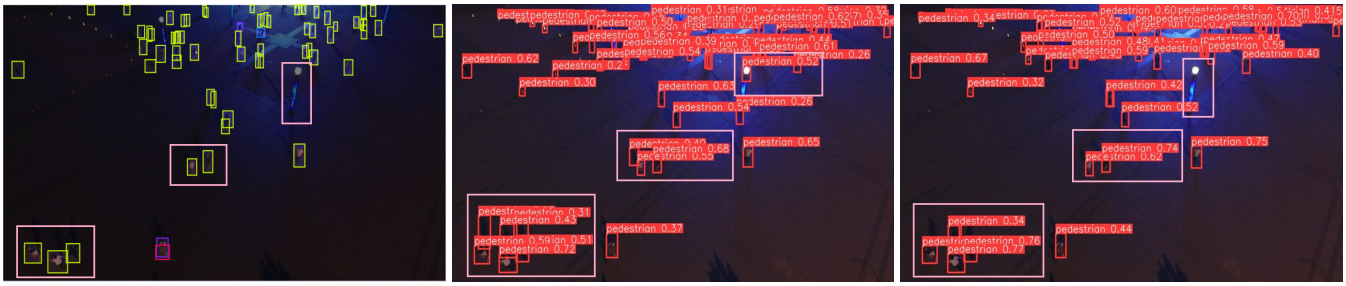
UAV image object detection faces numerous challenges. This section elaborates on how the improved YOLOv8n effectively addresses these issues.

To tackle the significant scale variation and often low image quality in UAV imagery, the model adopts three detection heads—Detect1 (40×40), Detect2 (80×80), and Detect3 (160×160)—each optimized for large, medium, and small object sizes, respectively. Detect3 focuses on small objects, Detect2 on medium-sized ones, and Detect1 on



(a) Original Image (b) YOLOv8n (c) Ours

FIGURE 10. Crowded street scene captured from above at dusk.



(a) Original Image (b) YOLOv8n (c) Ours

FIGURE 11. Crowded plaza at night captured from an oblique angle.

TABLE 8. Comparison with traditional object detection algorithms.

Network	mAP@0.5 (%)	mAP@0.5:0.95 (%)	Params (M)	GFLOPs
Faster R-CNN	37.2	21.9	41.16	292.8
Cascade R-CNN	39.1	24.3	68.95	320.7
RetinaNet	19.1	10.6	35.68	299.5
CenterNet	33.7	18.8	70.75	137.2
Ours	**42.1**	**27.7**	**2.3**	**12.0**

Bold font indicates the best values.

TABLE 9. Comparison among improved models.

Network	mAP@0.5 (%)	mAP@0.5:0.95 (%)	Params (M)	GFLOPs
LE-YOLO	39.3	22.7	2.1	13.1
SOD-YOLO-n	33.0	19.3	**0.6**	7.8
ESOD-YOLO	29.3	16.6	4.5	14.3
YOLOv5-pp	41.7	–	10.5	–
USV-YOLO	38.5	–	7.5	17.7
YOLO11n	33.2	19.3	2.6	6.3
PS-YOLO-n	33.3	19.3	1.39	**5.1**
TOOD	40.8	25.6	32	199
Ours	**42.1**	**27.7**	2.3	12.0

Bold font indicates the best values.

large targets. This multi-level design enables the model to effectively adapt to varying object scales and significantly improves small object detection. As shown in Table 2, the modified YOLOv8n demonstrates a clear advantage in detecting small targets.

To further strengthen the model's small object feature extraction capabilities, the LSKA attention mechanism is embedded into the SPPF module, significantly enhancing multi-scale feature fusion and the semantic representation of small targets. As shown in Table 3, the SPPF-LSKA module improves overall feature extraction performance.

In UAV imagery from the VisDrone2019 dataset, overlapping and occluded objects often lead to false positives and missed detections. To address this, we incorporate Soft-NMS

TABLE 10. Performance comparison on RSOD dataset.

Network	mAP@0.5(%)	Params (M)	GFLOPs	SIZE (MB)
YOLOv8n	93.7	3.00	**8.1**	6.2
YOLOv10n	93.4	2.69	8.2	5.8
Deformable DETR	91.0	40.0	196	–
RT-DETR-r18	93.9	19.8	57	76.9
Ours	**97.1**	**2.33**	12.0	**5.0**

Bold font indicates the best values.

and the MPDIoU loss function. Soft-NMS alleviates missed detections by adjusting confidence scores of overlapping boxes, avoiding the rigid suppression of traditional NMS. MPDIoU improves bounding box regression by considering geometric similarity, enabling more accurate localization of occluded and closely spaced objects. These enhancements notably improve detection reliability, even under low-light or densely populated conditions.

In summary, the improved YOLOv8n significantly boosts object detection accuracy in UAV images. As demonstrated in Figure 6 and Table 7, most categories saw a substantial increase in accuracy, with only a slight decrease observed in the "car" category. This confirms that the proposed improvements effectively address the unique challenges posed by drone-based object detection.

VI. CONCLUSION

This study presents an enhanced object detection model based on YOLOv8n, tailored for small object detection in UAV imagery and addressing typical challenges such as occlusion and scale variation. The model combines the SPPF and LSKA modules into a unified SPPF-LSKA structure to facilitate efficient multi-scale feature fusion. A 160×160 feature map is added to boost feature extraction capabilities. Additionally, the integration of Soft-NMS and the MPDIoU loss function improves the model's performance in detecting overlapping and occluded objects.

Experimental results on the VisDrone2019 dataset show that the improved model surpasses the baseline YOLOv8n, with mAP@0.5 and mAP@0.5:0.95 both increasing by 7.6%. Compared to other detection algorithms such as YOLOv11, it achieves an 8% gain in mAP@0.5, demonstrating its strong performance in UAV-based small object detection tasks.

It is worth noting that the current network improvements do not involve optimization of the head components, nor do they incorporate mainstream attention mechanisms or recent advances such as integrating Transformer-based structures into YOLOv8. In addition, no advanced image processing techniques were applied for the preprocessing of UAV imagery. The real-time performance of the algorithm has slightly decreased compared to the original version, and there is still room for further improvement in detection accuracy. These aspects leave ample space for future enhancements of the model.

Moreover, the improved YOLOv8n model has only been evaluated on the VisDrone2019 and RSOD datasets. To comprehensively assess its performance in real-world scenarios, further validation on a wider range of UAV platforms and more diverse UAV datasets is necessary. This will help promote improvements in the model's generalization ability and robustness.

DECLARATIONS

- Data availability: The datasets generated and analysed during the current study are available in the availability: https://github.com/666xkk/Small-Object-Detect-Project/tree/master. https://github.com/VisDrone/VisDrone-Dataset. https://github.com/RSIA-LIESMARS-WHU/RSOD-Dataset-

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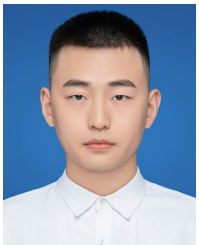

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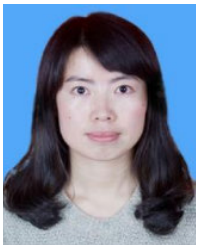

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